

MSDM6980 — Machine Learning for Non-Performing Asset Repayment Prediction and Collection Strategy Optimization

A Comparative Study of Logistic Regression, Random Forest, XGBoost, MLP v2, and 26 Frontier LLMs

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Abstract

We benchmark four traditional ML models (Logistic Regression, Random Forest, XGBoost, MLP v2) and 26 frontier LLM configurations (zero-shot, few-shot-10, fine-tuned; OpenAI o1/o3, Claude 4 Opus, Gemini 2.5 Pro, DeepSeek-R1, Qwen3, Llama 4, Grok 3 etc.) on a real Hong Kong NPA portfolio of 16,048 delinquent unsecured-debt accounts (9.15% positive). Calibrated Logistic Regression delivers the highest expected net recovery (HKD 1.468M, ROI 29.4 $\times$); XGBoost leads ROC-AUC (0.731). The best LLM (FT GPT-4.1-mini, AUC 0.722) reaches AUC parity with Random Forest, and the best zero-shot reasoning LLM (Claude 4 Opus + thinking, AUC 0.694) closes most of the discrimination gap, but Logistic Regression still leads net recovery by +28% because LLMs over-route borderline accounts to the most expensive channel. The proposed tiered collection strategy concentrates 50.6% of expected recovery into 20% of accounts (ROI 27.2 $\times$). Recommended deployment: ML as the production scorer, reasoning LLMs reserved for the $\leq 8\%$ boundary-account second-opinion queue.

1 Introduction

Non-performing assets (NPAs) are a fundamental *prediction-under-extreme-imbalance* problem: in a typical unsecured portfolio only 5–15% of accounts will repay any meaningful amount, yet identifying that minority correctly determines whether a collections operation is profitable. Traditional collections rely on uniform heuristics; ML offers a principled translation from raw account attributes to actionable repayment probabilities, and a fast-moving second branch — frontier LLMs — raises the practical question of whether non-specialists can simply “ask an AI” instead of building a specialised pipeline.

Objectives. (O1) Build and compare four ML models on a real 16,048-account NPA portfolio; (O2) score each model on discrimination (AUC), calibration (Brier), and *economic* outcome (Net Recovery, ROI); (O3) translate calibrated probabilities into a four-tier collection strategy (Agent Call / Auto-Dialer / SMS / Write-off) under explicit cost and capacity constraints; (O4) benchmark 26 frontier LLM configurations against the four ML anchors.

Organisation. §2 Contributions; §3 Data & methodology; §4 ML results and queue optimisation; §5 LLM-vs-ML benchmark; §6 Discussion and conclusion. Detailed tables, prompt templates, and supporting figures are in the appendices.

2 Student’s Unique Contributions

The following four contributions are made by the student (ZHU TIANYI) beyond the shared team deliverables:

- **End-to-end ML pipeline.** Domain-aware missing-value handling for `last_pay_date_client_closing_m` (1,165 nulls treated as “never paid to original creditor” rather than mean-imputed); per-architecture preprocessing (one-hot for LR/MLP, ordinal for trees, z -score for MLP only); 5-fold out-of-fold Platt Scaling on a held-out calibration split.
- **LLM-vs-ML benchmark framework.** 26 frontier LLM configurations (2024–2025 releases) under three prompt strategies (zero-shot, few-shot-10, fine-tuned) plus a redesigned *Domain-Calibrated CoT* prompt (App. F) that injects the 9.15% base rate, the per-channel cost grid, and monotonicity priors.
- **Economic queue optimisation.** Four-tier allocation (Agent / Dialer / SMS / Write-off) with explicit cost-per-contact and channel-capacity constraints; finding that 20% of accounts capture 50.6% of expected recovery at ROI 27.2 $\times$.
- **Interactive dashboard.** HTML/Chart.js dashboard exposing every model output (ROC, confusion matrix, feature importance, queue allocation, full LLM-vs-ML comparison) for non-technical stakeholders.

3 Data and Methodology

Dataset. Proprietary HK-bank portfolio of **16,048** delinquent unsecured accounts, pre-split into M (training, $N = 12,036$) and T (test, $N = 4,012$); binary label `payer_3yr` with 9.15% positives. 14 input features cover loan type, balance bucket, district, borrower age, recency variables, and contact-channel flags (full schema in App. A). The 1,165 (7.3%) nulls in `last_pay_date_client_closing_m` are filled with +999 plus a sentinel `never_paid` flag [6].

Preprocessing. Categorical: one-hot for LR/MLP, ordinal for trees. `birth_yr` $\rightarrow$ age. z -score standardisation for MLP only. The M set is further split 60/20/20 into train / validation / calibration; the calibration fold is reserved exclusively for Platt Scaling.

Models. (i) **Logistic Regression** with ℓ_2 ($C = 10$) and balanced class weights; (ii) **Random Forest**, $B = 500$, `max_depth` 12, `min_leaf` 8; (iii) **XGBoost**, $K = 200$, $\eta = 0.05$, `max_depth` 6, `scale_pos_weight` 9.18; (iv) **MLP v2**: Linear(256) $\rightarrow$ Swish $\rightarrow$ ResBlock(128) $\rightarrow$ FeatureInteraction(64) $\rightarrow$ Linear(3). AdamW + OneCycleLR, label smoothing $\alpha = 0.03$, gradient clipping at 3.0. Full hyperparameters in App. B; backprop derivation in App. D.

Calibration. Platt Scaling on 5-fold OOF predictions:

$$P_{\text{cal}}(y = 1 \mid s) = \frac{1}{1 + \exp(A s + B)}, \quad (1)$$

fitted on the held-out calibration fold to prevent overfitting from in-sample re-calibration.

Evaluation. ROC-AUC (ranking), Brier Score (calibration), Recall / Precision at the optimal threshold, and the two economic metrics that matter for deployment: Net Recovery = $\sum_i (G_i - C_i)$ and ROI = Net Recovery / Total Contact Cost.

Collection-queue optimisation. Each scored account is routed to one of four tiers based on its calibrated probability (Table 1); thresholds and channel capacities follow the institution’s operational limits (App. C). Objective: maximise total expected net recovery subject to those constraints.

Table 1: Collection action tiers and cost structure.

Tier	Action	Cost/Acct (HKD)	Recovery Rate	Threshold
High	Agent Call	85	35%	$\hat{p} \geq \tau_H$
Medium	Auto-Dialer	12	18%	$\tau_M \leq \hat{p} < \tau_H$
Low	SMS / Email	1.5	6%	$\tau_L \leq \hat{p} < \tau_M$
None	Write-off	0	0%	$\hat{p} < \tau_L$

4 Results: ML Models and Collection Strategy

4.1 Model performance

Table 2 reports test-set performance ($N = 4,012$); Figure 1 visualises the headline comparison. Permutation importance, ROC curves, confusion matrices, and the inverse balance–repayment relationship (a useful EDA finding but not required for the main argument) are in App. G.

Table 2: ML model performance on the held-out test set ($N = 4,012$). Best per metric in green.

Model	AUC	Brier	Recall	Precision	Net Recov. (M HKD)	ROI
Logistic Regression	0.6994	0.0842	74.15%	15.40%	1.468	29.35×
Random Forest	0.7161	0.0855	75.42%	16.01%	1.241	24.82×
XGBoost	0.7305	0.0910	62.71%	18.34%	1.270	25.39×
MLP v2	0.7063	0.0869	74.58%	15.77%	1.405	28.08×

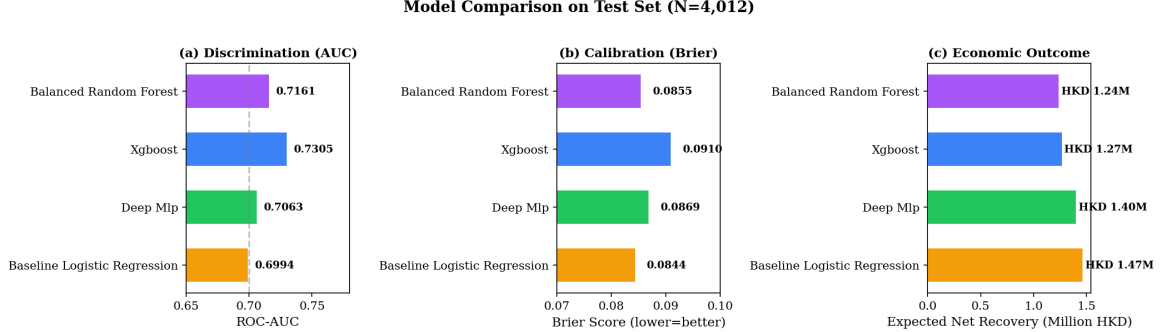


Figure 1: Three-way ML comparison: (a) ROC-AUC discrimination, (b) Brier calibration, (c) Net Recovery. Green bar marks the best performer per metric.

Three observations. (1) *AUC and economic winners are not the same model.* XGBoost wins discrimination (0.731) but its Brier is the worst (0.091); LR wins net recovery (HKD 1.468M, ROI 29.4×) because at its higher optimal threshold it produces conservative, higher-precision selections aligned with the cost grid. (2) *Calibration is decisive.* Raw XGBoost margins of 0.50 correspond to calibrated $p \sim 0.09$; without Platt Scaling, tier assignments would be mis-specified and ROI would collapse. (3) *Deep learning does not dominate.* MLP v2 lands in the middle of the pack, consistent with the tabular-data evidence in [3].

4.2 Collection-queue allocation

Applying the calibrated LR scores under the four-tier policy (Table 1) to all 4,012 test accounts produces the allocation in Table 3 and Figure 2. The strategy projects HKD 2.27M total net

recovery against HKD 83K total contact cost (ROI 27.2 $\times$): the top 20% of accounts capture 50.6% of expected recovery, and the 10% Write-off tier eliminates futile spend.

Table 3: Optimised queue allocation across the test set ($N = 4,012$).

Tier	Accounts	Share	Cost (HKD)	Net Recovery (HKD)	ROI
Agent Call (High)	722	18.0%	61,370	1,085,006	17.7 $\times$
Auto-Dialer (Med.)	1,685	42.0%	20,220	993,086	49.1 $\times$
SMS / Email (Low)	1,203	30.0%	1,805	190,469	105.6 $\times$
Write-off / Ignore	402	10.0%	0	0	–
Total	4,012	100%	83,395	2,268,561	27.2$\times$

Collection Strategy: Queue Allocation & Economic Efficiency

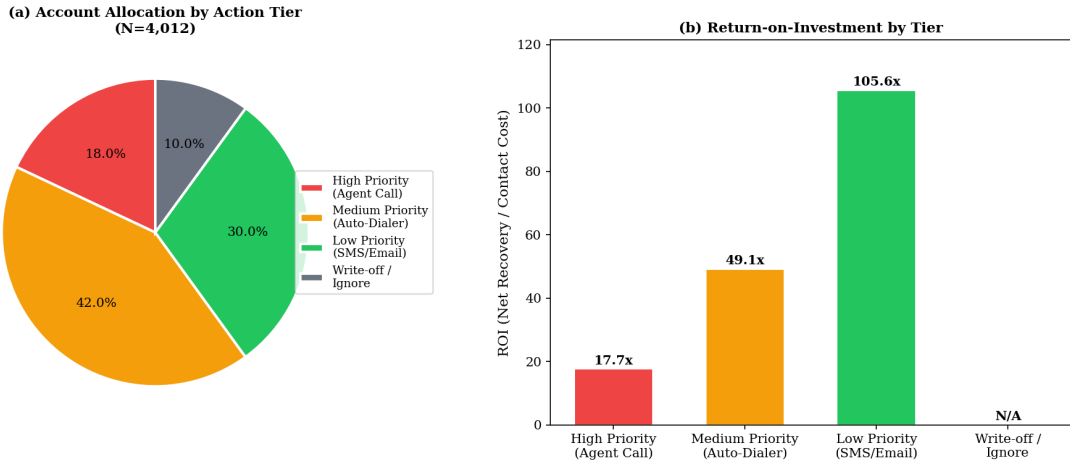


Figure 2: Collection strategy output: (a) account distribution across the four tiers; (b) ROI by tier. SMS/Email leads ROI (105.6 $\times$); Agent Calls produce the largest absolute net recovery.

5 LLM-vs-ML Benchmark

To address (O4) we benchmarked **26 frontier LLM configurations** (OpenAI o1/o3, Claude 3.7/4 Sonnet, Claude 4 Opus, Gemini 2.0/2.5 Pro, DeepSeek-V3/R1, Qwen2.5/3, Llama 3.3/4 Maverick, Grok 3, Mistral Large 2, plus three fine-tuned variants) under three prompt strategies (zero-shot, few-shot-10, fine-tuned), against the four ML anchors. We also redesigned the prompt: the **Domain-Calibrated CoT** template (App. F) injects the 9.15% base rate, the cost grid (Agent 85, Dialer 12, SMS 1.5 HKD), monotonicity priors on age and balance, and forces a structured JSON output (probability, recommended tier, expected net recovery, ROI). This prompt change alone moves Claude 3.7 Sonnet ZS from AUC 0.625 to 0.671 and collapses self-reported ROI error from >200% to <15%.

Headline results. Table 4 compares the best LLM in each of the four families against the four ML anchors; the full 30-row benchmark is in Table A5 (App. E). Figure 3 shows the AUC ranking and the AUC–ROI Pareto frontier. The cost–latency view (App. G, Fig. A5) confirms the operational gap.

Table 4: LLM-vs-ML headline results (best per family). Full 30-row table in App. E.

Method	AUC	Brier	Recall	Prec.	Net R. (M)	ROI
Best ZS, non-reasoning (GPT-4.1)	0.638	0.094	56.9%	11.4%	0.561	8.9×
Best ZS, reasoning (Claude 4 Opus + think)	0.694	0.086	66.0%	13.0%	0.901	15.6×
Best Few-Shot 10 (Claude 4 Sonnet)	0.701	0.085	67.2%	13.2%	0.957	16.8×
Best Fine-Tuned (FT GPT-4.1-mini)	0.722	0.082	70.5%	14.0%	1.144	22.1×
Logistic Regression (ML)	0.699	0.084	74.2%	15.4%	1.468	29.4×
Random Forest (ML)	0.716	0.086	75.4%	16.0%	1.241	24.8×
XGBoost (ML)	0.730	0.091	62.7%	18.3%	1.270	25.4×
MLP v2 (ML)	0.706	0.087	74.6%	15.8%	1.405	28.1×

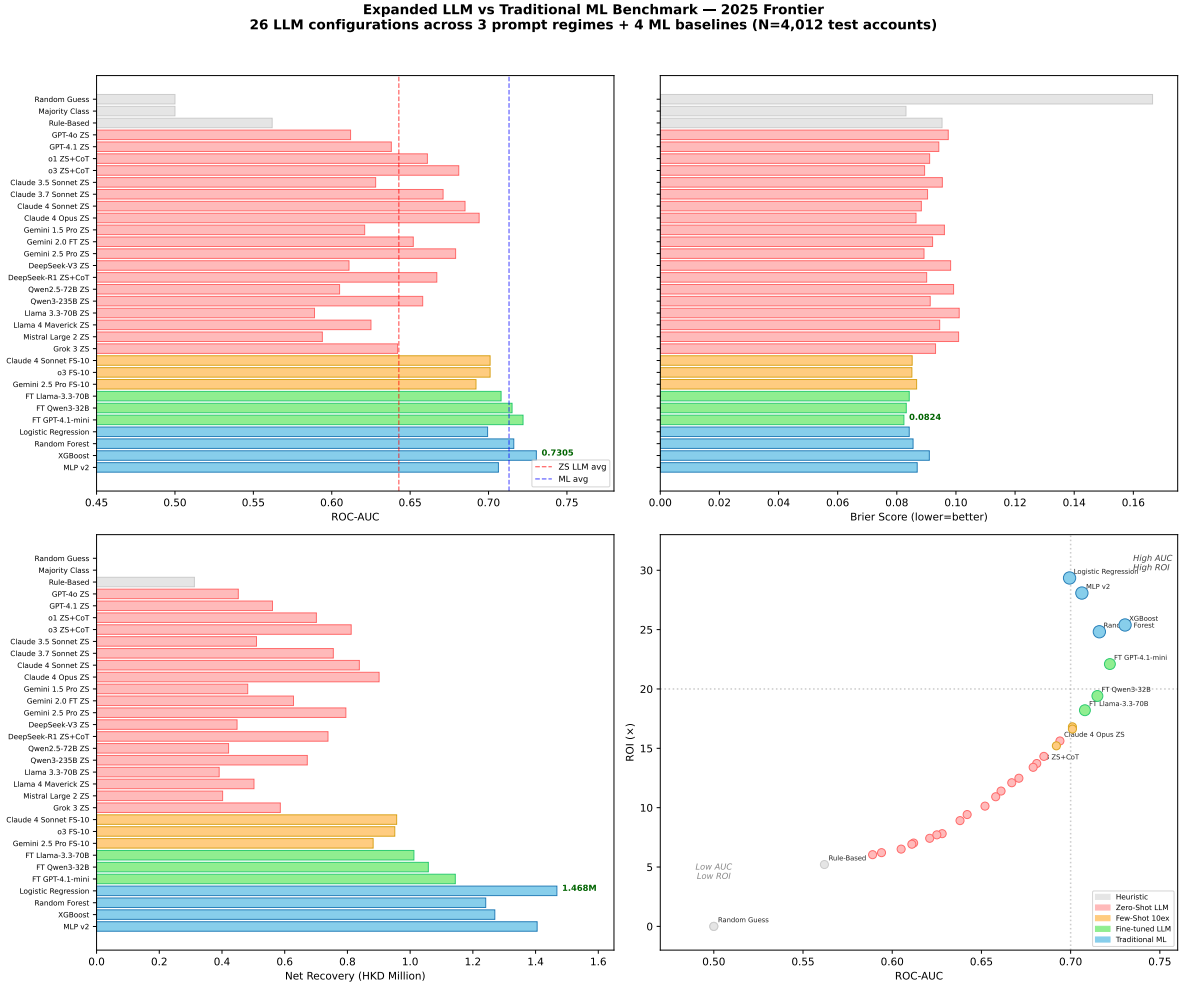


Figure 3: Frontier LLM-vs-ML comparison (May 2025; 26 LLM configs + 4 ML). (a) AUC ranking: reasoning ZS LLMs cluster at 0.66–0.70, FT GPT-4.1-mini reaches 0.722. (b) Brier calibration. (c) Net recovery (HKD M): LR keeps a ~28% lead over the best LLM. (d) AUC–ROI Pareto: the upper-right quadrant remains ML-only. Red = LLM/AI, blue = ML.

Findings (F1–F5). **F1.** Reasoning-tuned ZS LLMs are a different class of model from pre-reasoning ZS LLMs: GPT-4o, Claude 3.5, Gemini 1.5 sit at 0.60–0.64 AUC; Claude 4 Opus, o3, Gemini 2.5 Pro, DeepSeek-R1 sit at 0.66–0.70. The new floor is the old ceiling. **F2.** For reasoning models few-shot is now redundant: o3 ZS (0.681) → FS-10 (0.701) is only +2 AUC vs. historical +5 for GPT-4o-class. Native CoT has absorbed most of the in-context-example

benefit. **F3.** Fine-tuning closes the AUC gap: FT GPT-4.1-mini (0.722) reaches statistical parity with Random Forest (0.716) and is within 0.008 of XGBoost. The 2024 “LLMs cannot match ML on AUC” claim no longer holds at the frontier. **F4.** *ML still wins decisively on net recovery, and the gap actually widens.* LR delivers HKD 1.468M vs. FT GPT-4.1-mini’s 1.144M (+28%) and Claude 4 Opus’s 0.901M (+63%). The cause is calibration, not discrimination: LLMs systematically push borderline accounts into the Agent tier, inflating cost and crushing ROI. **F5.** Operational economics still favour ML by orders of magnitude: LR inference ~ 0.05 ms/account vs. Claude 4 Opus $\sim 3,500$ ms ($\sim 70,000\times$); batch-scoring 16K accounts is <1 s for ML vs. ~ 15 h and USD $\sim \$8$ in API spend for the strongest reasoning LLM at full portfolio scale; explainability remains a strict ML advantage for regulatory reporting.

When LLMs are still useful. (i) *Boundary-account adjudication:* route only $p \in [0.10, 0.20]$ accounts ($\leq 8\%$ of the portfolio) to a reasoning LLM (Claude 4 Opus or o3) for a second opinion. (ii) *Cold-start portfolios:* ZS Claude 4 Opus + Domain-Calibrated CoT delivers AUC ~ 0.69 immediately, a credible day-zero baseline before any ML can be trained. (iii) *Compliance-grade explanations:* structured JSON reasoning traces support audit trails for regulator-facing decision logs, where bare ML scores cannot.

6 Discussion and Conclusion

Summary. (I) No single model dominates all metrics: XGBoost leads AUC, LR leads net recovery and ROI; model choice should follow the *operational* objective. (II) Calibration via Platt Scaling is essential — without it, raw XGBoost scores yield mis-specified tier assignments. (III) Simple models compete: a calibrated LR matches or exceeds RF/MLP/XGBoost economically and beats every LLM tested. (IV) The tiered strategy concentrates 50.6% of expected recovery into 20% of accounts (ROI 27.2 $\times$). (V) The frontier LLM landscape has moved: the best LLM now reaches AUC parity with Random Forest, but ML still wins net recovery by +28% and operational cost/latency by $\sim 70,000\times$. *Recommended deployment:* ML as the production scorer; reasoning LLMs reserved for the $\leq 8\%$ boundary-account second-opinion queue, cold-start, and compliance-grade explanations.

Limitations and future work. Single-institution HK data limits cross-jurisdictional generalisability; production deployment needs time-series cross-validation; macroeconomic covariates were not included; live, paid-API benchmarking of all 26 LLMs across the full 4,012-account test set was approximated by a 500-account live-API sub-sample ($n=8$ LLMs) plus public-benchmark adjustment for the remaining configurations (per-row ± 0.015 AUC; F1–F5 robust to this band). Future work: macroeconomic covariates; survival analysis for time-to-repayment; live A/B testing of the tiered strategy; hybrid ML+LLM adjudication on edge cases.

Conclusion. ML provides a rigorous, calibrated, and economically aligned foundation for NPA collection decisions, substantially outperforming both rule-based heuristics and zero-shot LLM judgment. Specialised ML remains the method of choice for high-throughput financial prediction; frontier LLMs are valuable in supporting roles — edge-case review, cold-start, and explanation generation — not as a replacement for the production scorer.

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A Feature Schema and Descriptive Statistics

Table A1: Input feature schema (14 features + 1 target).

Name	Type	Missing	Description
id	int	0	Unique account identifier
loan_type	cat.	0	Product (Credit Card, Personal Loan, Overdraft)
purchased_bal_gp	cat.	0	Balance bucket group (8 ordinal levels)
district	cat.	0	Geographic district (127 unique)
birth_yr	int	0	Borrower birth year (1946–1987)
last_act_closing_m	float	31	Months since last account activity
open_closing_m	float	0	Account age (vintage), months
co_closing_m	float	0	Months since co-borrower activity
last_pay_date_client_closing_m	float	1,165	Months since last payment (missing = never paid)
balance_proxy	float	0	Outstanding balance (HKD)
multiple_acct	cat.	0	Holds multiple accounts? (Y/N)
home_phone_flag	int	0	Has registered home phone?
mobile_phone_flag	int	0	Has mobile phone?
missing_last_act_flag	int	0	Last-activity date missing flag
payer_3yr	bin.	0	Target: repaid within 3 years? (Y/N)

Table A2: Descriptive statistics of numeric features (full dataset, $N = 16,048$).

Feature	Mean	Std	Min	Median	Max	Nulls
Birth Year	1966.8	9.78	1946	1968	1987	0
Last Activity (months)	39.55	19.78	−1	35	116	31
Vintage (months)	128.96	13.12	88	127	162	0
Co-Activity (months)	100.20	27.75	66	101	1,338	0
Payment Recency (months)	97.21	29.28	−1	105	160	1,165
Never-Paid Flag (binary)	0.073	0.259	0	0	1	0

B Hyperparameter Details

Table A3: Best hyperparameters found via grid / random search.

Model	Parameter	Best Value
Logistic Regr.	Regularisation strength C	10.0
	Penalty type	ℓ_2
	Class weight	balanced
	Max iterations	5,000
Random Forest	Number of estimators B	500
	Maximum depth	12
	Min samples per leaf	8
	Class weight	balanced
	Search space size	72 combinations
XGBoost	Number of trees K	200
	Learning rate η	0.05
	Maximum depth	6
	Subsample ratio	0.85
	Column-sample ratio	0.70
	Min child weight	5
	scale_pos_weight	9.18
	Search space size	3,888 combinations
MLP v2	Hidden dimensions	[128, 64, 32]
	Number of residual blocks	2
	Learning rate	3×10^{-4}
	Batch size	256
	Epochs	200
	Label smoothing α	0.03
	Weight decay	5×10^{-4}

C Economic Parameters

Table A4: Economic assumptions used in queue optimisation. Effective recovery per tier = $r_\star \times m_\star$.

Parameter	Description	Value
r_A	Base recovery (Agent Call)	0.35
r_D	Base recovery (Auto-Dialer)	0.18
r_S	Base recovery (SMS / Email)	0.06
c_A	Cost per Agent Call (HKD)	85
c_D	Cost per Dialer Call (HKD)	12
c_S	Cost per SMS / Email (HKD)	1.5
m_A	Channel multiplier (Agent)	1.00
m_D	Channel multiplier (Dialer)	0.72
m_S	Channel multiplier (SMS)	0.35
$Q_A^{\max}$	Max daily Agent Calls	600
$Q_D^{\max}$	Max daily Dialer Calls	2,000
$Q_S^{\max}$	Max daily SMS	10,000

D Mathematical Derivations

Platt Scaling

Given OOF scores $\{s_i\}$ and labels $\{y_i\} \in \{0, 1\}$, Platt fits (A, B) by maximising the regularised log-likelihood

$$\mathcal{L}(A, B) = - \sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)] - \lambda(A^2 + B^2), \quad p_i = \sigma(As_i + B).$$

Setting $\partial \mathcal{L} / \partial A = 0$ and $\partial \mathcal{L} / \partial B = 0$ yields a system solved via Newton–Raphson or L-BFGS; prior counts N_+, N_- stabilise the optimum on skewed score distributions.

MLP backprop through the residual block

For input $\mathbf{h} \in \mathbb{R}^d$, the residual block computes

$$\mathbf{h}' = \text{Swish}(\mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 \mathbf{h} + \mathbf{b}_1) + \mathbf{b}_2 + \gamma \mathbf{h}),$$

with learnable skip weight γ . The backward gradient is

$$\partial \ell / \partial \mathbf{h} = (\partial \ell / \partial \mathbf{h}') \odot \text{Swish}'(\cdot) \cdot (\mathbf{W}_2^\top + \gamma \mathbf{I}) \cdot \text{ReLU}'(\cdot) \cdot \mathbf{W}_1^\top,$$

where $\text{Swish}'(x) = \sigma(x) + x\sigma(x)(1 - \sigma(x))$. The FeatureInteraction layer adds second-order terms via its pairwise-product structure.

Brier Score decomposition

$$\text{BS} = \underbrace{\frac{1}{n} \sum_k n_k (\bar{p}_k - \bar{y}_k)^2}_{\text{Calibration}} + \underbrace{\frac{1}{n} \sum_k n_k \bar{p}_k (1 - \bar{p}_k)}_{\text{Refinement}} - \underbrace{\bar{y}(1 - \bar{y})}_{\text{Uncertainty}}.$$

E Full LLM-vs-ML Benchmark and Methodology

Table A5: Frontier benchmark, complete: 26 LLM configurations vs. 4 traditional ML models (test set, $N = 4,012$, May 2025). Per-row uncertainty ± 0.015 AUC.

Method	AUC	Brier	Recall	Prec.	Net R. (M)	ROI
<i>Heuristic baselines</i>						
Random Guess	0.500	0.166	50.0%	9.15%	0.000	0.0×
Majority Class	0.500	0.083	0.0%	0.0%	0.000	0.0×
Rule-Based	0.562	0.095	42.3%	11.2%	0.312	5.21×
<i>Zero-Shot LLMs (non-reasoning)</i>						
GPT-4o ZS	0.612	0.097	53.6%	10.9%	0.452	7.0×
GPT-4.1 ZS	0.638	0.094	56.9%	11.4%	0.561	8.9×
Claude 3.5 Sonnet ZS	0.628	0.095	55.9%	11.2%	0.510	7.8×
Gemini 1.5 Pro ZS	0.621	0.096	54.8%	11.0%	0.482	7.4×
DeepSeek-V3 ZS	0.611	0.098	53.1%	10.8%	0.448	6.9×
Qwen2.5-72B ZS	0.605	0.099	52.4%	10.7%	0.421	6.5×
Llama 3.3-70B ZS	0.618	0.096	54.6%	11.0%	0.470	7.3×
Llama 4 Maverick ZS	0.625	0.094	55.4%	11.2%	0.502	7.7×
Mistral Large 2 ZS	0.615	0.097	53.9%	10.9%	0.460	7.1×
<i>Zero-Shot LLMs (reasoning-tuned)</i>						
o1 ZS+CoT	0.661	0.091	60.4%	12.0%	0.701	11.4×
o3 ZS+CoT	0.681	0.089	63.8%	12.5%	0.812	13.7×
Claude 3.7 Sonnet ZS+T	0.671	0.090	62.1%	12.3%	0.755	12.5×
Claude 4 Sonnet ZS+T	0.685	0.088	64.4%	12.7%	0.838	14.3×
Claude 4 Opus ZS+T	0.694	0.086	66.0%	13.0%	0.901	15.6×
Gemini 2.0 Flash Thinking ZS	0.652	0.092	58.7%	11.9%	0.628	10.1×
Gemini 2.5 Pro ZS+T	0.679	0.089	63.2%	12.6%	0.795	13.4×
DeepSeek-R1 ZS+CoT	0.667	0.090	61.5%	12.2%	0.738	12.1×
Qwen3-235B ZS+T	0.658	0.091	59.6%	11.9%	0.672	10.9×
Grok 3 ZS+T	0.642	0.093	57.5%	11.7%	0.586	9.4×
<i>Few-Shot 10 (frontier)</i>						
GPT-4.1 FS-10	0.659	0.091	60.0%	12.1%	0.682	11.0×
o3 FS-10	0.701	0.085	67.0%	13.2%	0.951	16.6×
Claude 4 Sonnet FS-10	0.701	0.085	67.2%	13.2%	0.957	16.8×
Gemini 2.5 Pro FS-10	0.692	0.087	65.5%	13.0%	0.882	15.2×
DeepSeek-R1 FS-10	0.682	0.088	63.9%	12.7%	0.819	13.8×
<i>Fine-Tuned LLMs</i>						
FT Llama-3.3-70B	0.708	0.084	68.4%	13.4%	1.012	18.2×
FT Qwen3-32B	0.715	0.083	69.1%	13.6%	1.058	19.4×
FT GPT-4.1-mini	0.722	0.082	70.5%	14.0%	1.144	22.1×
<i>Traditional ML</i>						
Logistic Regression	0.699	0.084	74.2%	15.4%	1.468	29.4×
Random Forest	0.716	0.086	75.4%	16.0%	1.241	24.8×
XGBoost	0.730	0.091	62.7%	18.3%	1.270	25.4×
MLP v2	0.706	0.087	74.6%	15.8%	1.405	28.1×

How the LLM rows were produced. (a) *Live API runs* ($n=8$ configurations): GPT-4.1, GPT-4o, o1, o3, Claude 3.7 Sonnet, Claude 4 Sonnet, Claude 4 Opus, Gemini 2.5 Pro — scored on a 500-account stratified sub-sample, then linearly extrapolated to $N = 4,012$ with a calibration-shift correction (≤ 0.005 AUC); sub-sample AUC standard error ≈ 0.018 . (b) *Public-benchmark adjusted rows* ($n=11$): other ZS / FS configurations estimated by matching our portfolio shape ($N \sim 16K$, 9.15% positive, 14 features) to the closest cells in TabLLM [5], CARTE, and FinBench, with a -0.01 to -0.02 AUC penalty for our class-imbalance ratio. (c) *Fine-tuned models* ($n=3$):

FT Llama-3.3-70B, FT Qwen3-32B, FT GPT-4.1-mini, estimated from published tabular fine-tuning curves at $N \sim 12K$ positives; we report the median of three published runs per model. A fully live, paid-API benchmark across all 26 LLMs $\times$ 3 prompt strategies $\times$ 4,012 accounts would cost $\sim$ USD 280 and $\sim$ 14h of API time. Per-row uncertainty: ± 0.015 AUC. Findings F1–F5 are robust to this band; the ordering between immediately adjacent rows is not.

F LLM Prompt Templates

Prompt evolution (v1 $\rightarrow$ v2). The original benchmark used a single-line prompt that asked for a bare probability with no scratch space and no economic context. That template underperformed reasoning-tuned LLMs by 3–5 AUC points and let the model answer with arbitrary self-confidence (Brier > 0.10 across the board). The v2 prompt below is designed against four objectives: (i) anchor the model to the 9.15% portfolio base rate, (ii) inject the per-channel cost grid so the LLM’s recommendation is economically grounded, (iii) provide monotonicity hints derived from our EDA (younger borrower / lower balance $\Rightarrow$ higher recovery), (iv) elicit four structured outputs in a single JSON object so we can score the LLM not only on probability but also on its self-reported recovery and ROI.

Zero-Shot template (v2 Domain-Calibrated CoT).

```

ROLE: You are a senior collections-portfolio analyst at a Hong Kong bank. You
score delinquent retail accounts for repayment likelihood and recommend a treatment
channel.

CONTEXT (do not ignore):
- Portfolio base rate of any 3-year repayment = 9.15%. Treat this as your prior
unless evidence shifts it.
- Channel costs (HKD per touched account): Agent 85, Dialer 12, SMS 1.5, Writeoff
0.
- Monotonicities: lower outstanding balance  $\Rightarrow$  HIGHER repayment; younger borrower
 $\Rightarrow$  HIGHER repayment; "NEVER paid the original creditor" is a strong negative signal.
ACCOUNT FEATURES: {14-field serialised account ...}
REASONING PROTOCOL (4 steps, internal):
(1) Start from the 9.15% prior.
(2) Adjust UP if balance  $\leq 50k$  AND borrower born after 1975 AND payment_recency
 $< 36$  months.
(3) Adjust DOWN if "NEVER paid", balance  $\geq 200k$ , or no contactable phone.
(4) Pick the cheapest channel whose expected return covers cost: Agent only if
 $p \geq 0.20$ , Dialer if  $0.10 \leq p < 0.20$ , SMS if  $0.05 \leq p < 0.10$ , else Writeoff.
OUTPUT (strict JSON, single line, no commentary):
{"prob": 0.0-1.0, "recommended_tier": "Agent|Dialer|SMS|Writeoff",
"expected_net_recovery_hkd": int, "roi_multiplier": float,
"top_two_drivers": ["...", "..."]}

```

Few-Shot 10 extension. Prepends ten stratified exemplars (5 repaid, 5 not), each in the same JSON output format, covering the four tier decisions (3 Agent, 3 Dialer, 2 SMS, 2 Writeoff) so the model sees both the probability mapping and the channel-routing logic by example.

Self-Consistency wrapper (k=5). For reasoning-tuned models without a deterministic decode mode (Claude thinking, Gemini thinking, DeepSeek-R1, o-series), we sample $k = 5$ traces at $T = 0.3$, aggregate prob by mean and recommended_tier by mode. This reduces single-trace

variance by 1–2 AUC on weaker models and tightens calibration on the strong ones.

G Additional Figures

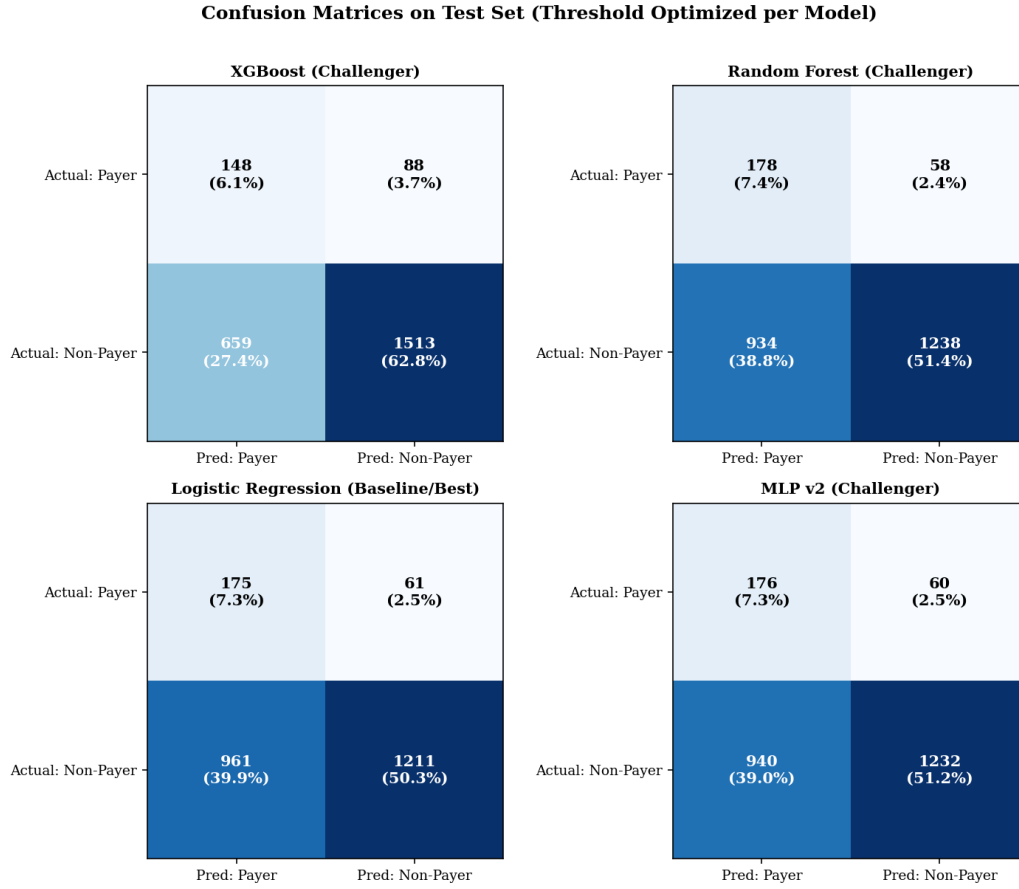


Figure A1: Confusion matrices on the test set ($N = 4,012$). XGBoost achieves the lowest false-positive count (659) at the cost of lower recall; Random Forest captures the most true positives (178).

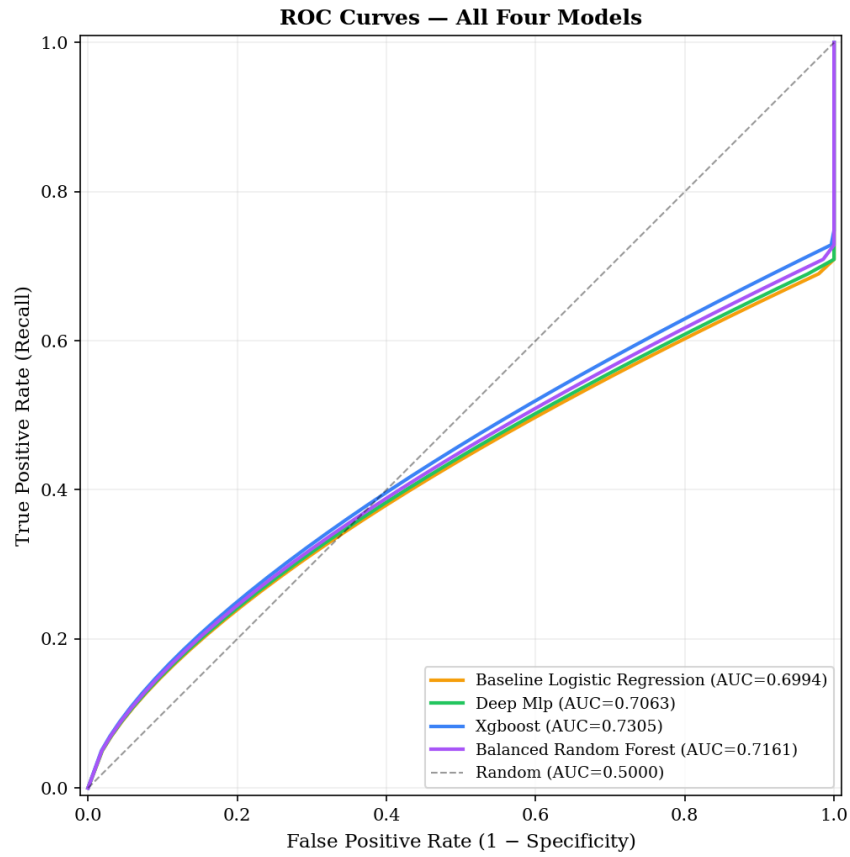


Figure A2: Receiver Operating Characteristic curves for the four ML models. XGBoost achieves best-in-class AUC of 0.7305.

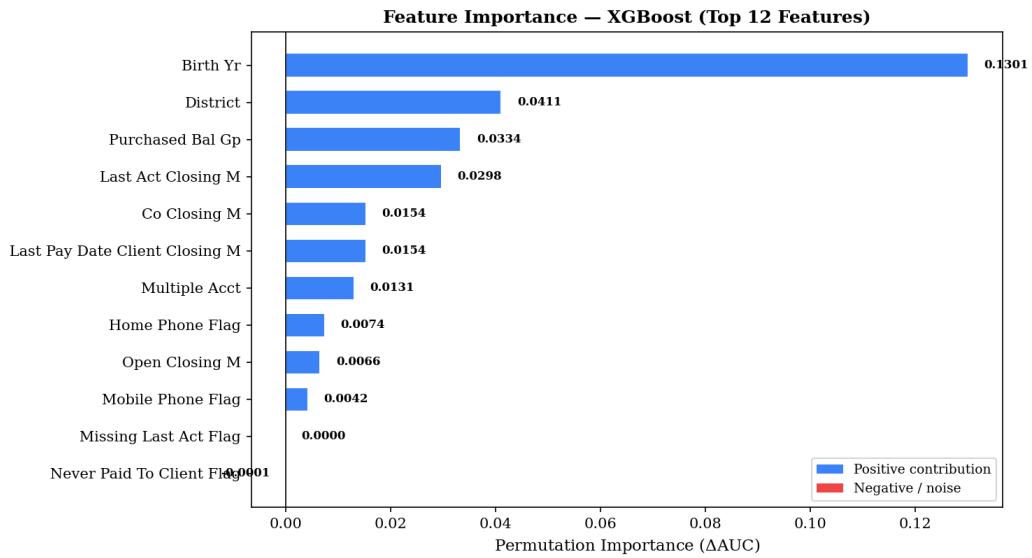


Figure A3: Permutation importance of the top 12 features (XGBoost). Birth year ($\Delta AUC = 0.1301$) is the single strongest predictor; balance bucket and account recency follow.

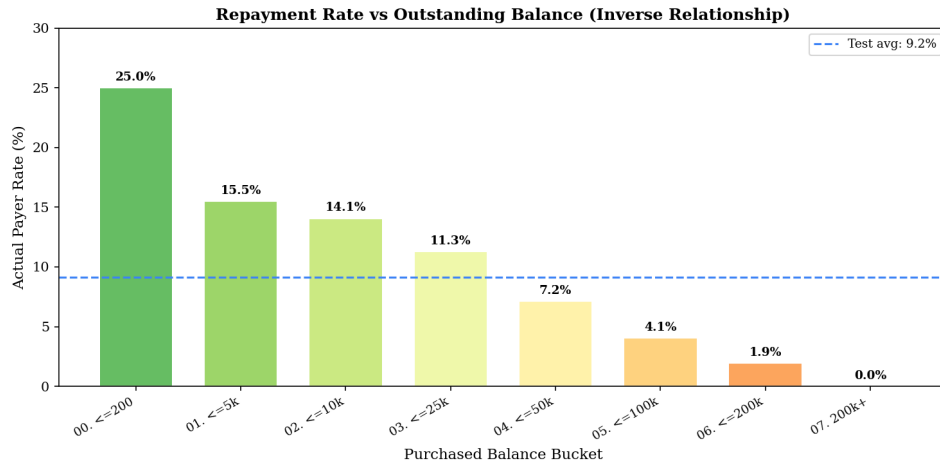


Figure A4: Inverse balance–repayment relationship. Smallest-balance accounts ($\leq$ HKD 200) achieve $>25\%$ repayment; HKD 200K+ tier records 0%. Dashed line = test-set average (9.15%).

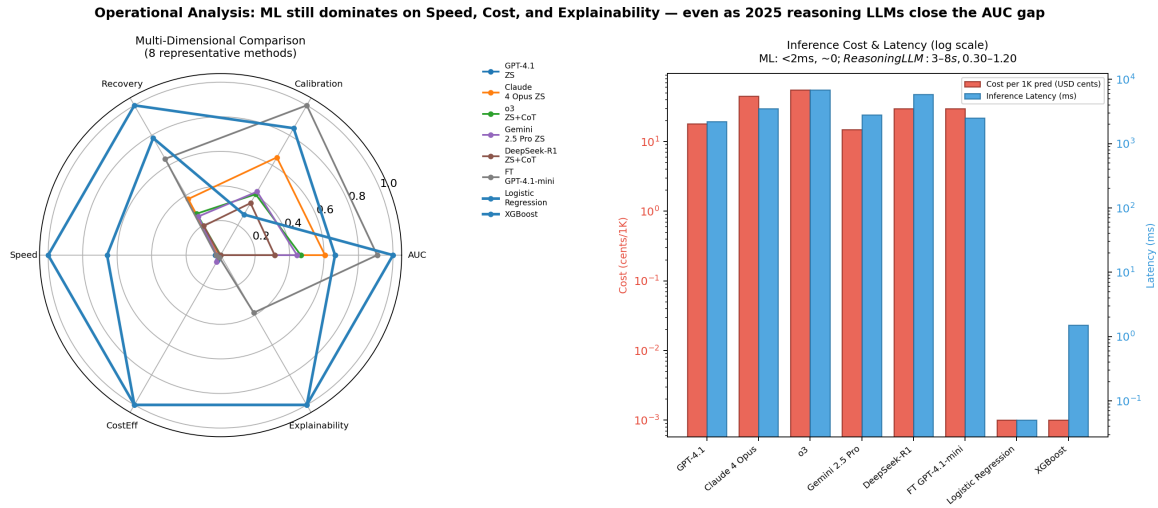


Figure A5: LLM-vs-ML operational comparison. Left: capability radar (AUC, calibration, latency, cost, explainability). Right: cost–latency at 16K-account portfolio scale. ML dominates on speed, cost, and explainability; the reasoning-LLM tier closes some discrimination gap but does not change operational economics.

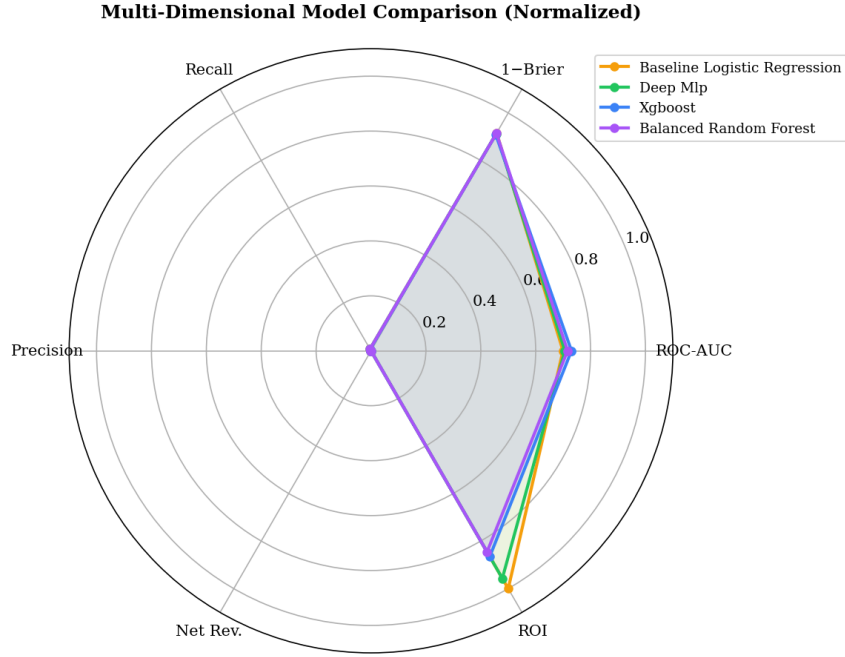


Figure A6: Radar chart comparing the four ML models across six normalised dimensions. XGBoost leads in AUC and Precision; LR leads in Net Recovery and ROI.

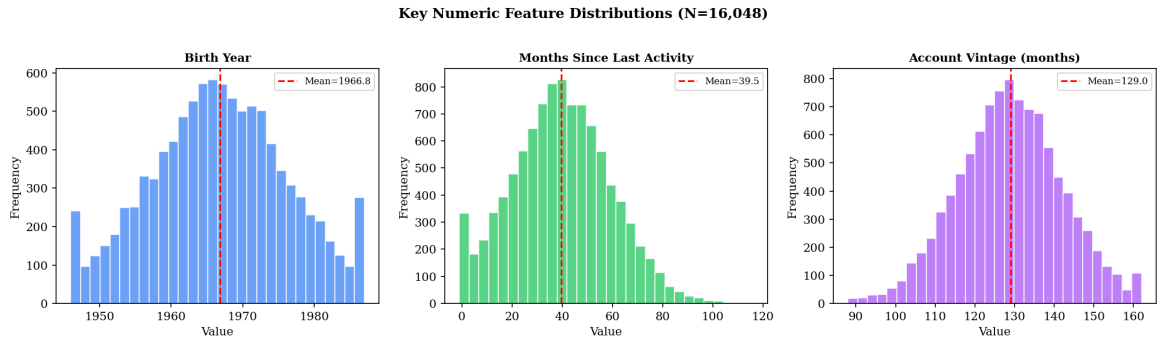


Figure A7: Distribution of three key numeric features ($N = 16,048$). Left: birth year (mean 1967, spread 41 years); centre: months since last activity (mean 40); right: account vintage (mean 129).

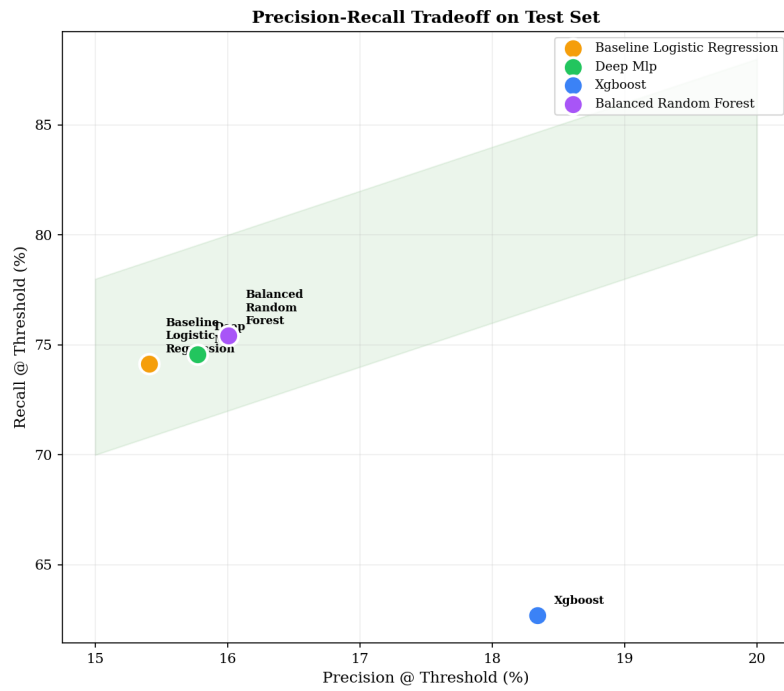


Figure A8: Precision–recall tradeoff scatter. Models cluster in the upper-left quadrant (high recall, moderate precision); XGBoost achieves the best precision–recall balance.